

The Common Overlap of Three Disks: an Exact, Backward-stable and Fast Solution for Mutual Transits and Eclipses

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ABSTRACT

The area common to three disks determines the light blocked during mutual events of transiting planets, moons and eclipsing multiple stars. The published closed forms are incomplete or incorrect: the formula of Fewell and the light-curve code LUNA that adopted it fail whenever the overlap contains more than half of one disk, a case later repaired by a dedicated branch, and the textbook lens formula used for the lens topologies loses all significant digits for thin overlaps. We present a new solution that removes every case distinction from the evaluation. We prove that a non-degenerate overlap has at most four corners, which reduces the geometry to five topologies resolved by a single directed walk along the boundary arcs. Green’s theorem then yields the area as a sum of circular segments and a vertex polygon whose evaluation requires one arctangent per arc, contains no subtractive cancellation, and treats arcs longer than π without a special case. Against 50-digit references for 21,568 configurations, including triangles of 10^{-15} times the smallest disk and lenses of depth 10^{-7} , the error never exceeds 2.01 times the condition number of the problem times the unit roundoff: the algorithm is componentwise backward stable, whereas the classical formulas exceed this bound by factors of up to 10^{16} . Five public packages that compute the same area, among them the mutual-transit code **gefera**, fail on 121 to 21,568 of 40,000 test configurations through undefined values, missing topologies or lost precision; the present method fails on none. The complete computation costs 45 ns per configuration on one CPU core, comparable to the bare closed form of a single topology and four to several thousand times faster than the public packages. The same decomposition delivers the centroid of the overlap in closed form. An open-source implementation is available at <https://github.com/user/repo>.

Keywords: Exoplanets — Transits — Exomoons — Eclipsing binary stars — Computational methods

1. INTRODUCTION

When two bodies transit a star simultaneously and overlap on the sky, the stellar light they block is the area of the union of their shadows on the stellar disk. With the star, a planet and its moon represented by three disks, the union follows from the pairwise lenses and one further quantity, the area of the common overlap of all three disks. Mutual events of this kind constrain the orbits of exomoons (J. Cabrera & J. Schneider 2007; D. M. Kipping 2011; M. Hippke & R. Heller 2022; S. Kálmán et al. 2023), planet–planet occultations (R. Luger et al. 2017) and multiply eclipsing stellar systems (D. R. Short et al. 2018). The single-body occultation is solved analytically (K. Mandel & E. Agol 2002; E. Agol

et al. 2020; R. Luger et al. 2019); the geometry of three disks underlies every analytic treatment of the mutual case.

M. P. Fewell (2006) gave the first closed form for the common overlap of three circles. He decomposed the circular triangle into the straight triangle on its three chords, evaluated with Heron’s formula, and three circular segments, and he classified the configurations into nine cases, of which only one is a circular triangle. His expression for a segment that exceeds half of its disk (his Eq. 16) retains the arcsine of the chord, $r^2 \arcsin(c/2r)$, which measures the minor arc, and omits the term πr^2 . D. M. Kipping (2011) transcribed Fewell’s construction to the star–planet–moon problem in the algorithm LUNA, organised the configurations into 27 principal cases with subcases, and inherited the error in his Eq. 37 (case 14.1b). T. A. Gordon & E. Agol (2023) identified the discrepancy between LUNA and an exact

computation near the inner edge of the stellar limb and published the corrected segment, $r^2[\pi - \arcsin(c/2r)] + (c/4)\sqrt{4r^2 - c^2}$, in their Appendix B. The root cause is structural: a chord length determines an arc only up to the choice between the arc and its complement, and every formula built on chord lengths requires an additional geometric test to make that choice.

The exact limb-darkened light curve of two mutually transiting bodies follows from Green’s theorem, which converts the flux integral over the visible stellar disk into line integrals along its boundary arcs (A. Pál 2012). T. A. Gordon & E. Agol (2023) implemented this approach in the code `gefera` with a tree of 16 geometric cases. Its published Fortran returns undefined values when an intersection of the two occulting bodies lies exactly on the stellar limb, and for a single body when $b + r = 1$, where the elliptic modulus reaches unity (Section 3.3). D. R. Short et al. (2018) integrate the line integrals numerically, R. Luger et al. (2017) discretise the star into annuli, and S. Kálmán et al. (2023) combine pixel classification with Gauss–Legendre quadrature in the Transit and Light Curve Modeller (S. Csizmadia 2020). Outside astronomy, F. Librino et al. (2014) computed intersections of many circles by inclusion–exclusion over a trellis of pairwise terms and confirmed Fewell’s construction as the reference for three circles. Area-proportional Euler and Venn diagrams require the same quantity: `eulerr` (J. Larsson 2026), `venn.js` (B. Frederickson 2018) and `matplotlib-venn` (K. Tretyakov 2025) compute intersections of circles from their crossing points, polygon areas and circular segments, and general geometry libraries such as Shapely (S. Gillies et al. 2025) on GEOS (GEOS contributors 2025) approximate disks by polygons. LUNA is not public.

All of these approaches rest on the area of a two-disk lens. Its textbook form (E. W. Weisstein 2024),

$$L = r_1^2 \arccos \frac{d^2 + r_1^2 - r_2^2}{2dr_1} + r_2^2 \arccos \frac{d^2 + r_2^2 - r_1^2}{2dr_2} - \frac{1}{2}\sqrt{\Pi}, \quad (1)$$

with $\Pi = (-d+r_1+r_2)(d+r_1-r_2)(d-r_1+r_2)(d+r_1-r_2)$, subtracts quantities of nearly equal size when the lens is thin, and the arccosine of an argument $1 - \epsilon$ amplifies the rounding error of the argument by $\epsilon^{-1/2}$. For a small disk that barely enters a large one, the regime of a moon at stellar ingress, Equation (1) loses every significant digit (Section 3.1). The same cancellation affects the intersection points computed as $h^2 = r_1^2 - a^2$, which enter all vertex-based formulas.

This paper presents a solution that is free of case-dependent formulas, correct for arcs of any length, backward stable in floating-point arithmetic, and as fast as the closed form of a single topology. Section 2 derives the method, Section 3 quantifies its accuracy and cost, and Section 4 discusses its scope.

2. METHOD

2.1. Topology

Let D_k denote the closed disk of centre $\mathbf{o}_k$ and radius r_k bounded by the circle C_k , $k = 0, 1, 2$, and let $R = D_0 \cap D_1 \cap D_2$. If $D_i \subseteq D_j$ for some $i \neq j$, then $D_i \cap D_j = D_i$ and D_j is removed; one remaining disk gives $A = \pi r^2$ and two give the lens of the remaining pair. If any pair is disjoint, $A = 0$. In the remaining configurations every pair of circles crosses at two points. A crossing of C_i and C_j that lies strictly inside D_k is a vertex of R ; let V be their number.

Proposition 1. *If no crossing lies on the third circle, then $V \in \{0, 2, 3, 4\}$. The overlap is empty for $V = 0$, the lens of one pair for $V = 2$, a circular triangle with one vertex from each pair for $V = 3$, and a circular quadrilateral whose four vertices lie on one circle for $V = 4$.*

Proof. R is convex. If its interior is non-empty and R is not a disk, its boundary is a simple closed curve composed of a_k arcs of each C_k , joined at the vertices, so that $V = a_0 + a_1 + a_2$. An arc of ∂R on C_i is a component of $(C_i \cap D_j) \cap (C_i \cap D_k)$, the intersection of two proper arcs of C_i ; hence $a_i \leq 2$, and $a_i = 2$ holds exactly when the two arcs cover C_i , i.e., $C_i \subset D_j \cup D_k$. Because $D_j \cap D_k \neq \emptyset$, the union $D_j \cup D_k$ is simply connected, and by the Jordan curve theorem it then contains D_i . A crossing $\mathbf{q}$ of C_j and C_k lies on the boundary of $D_j \cup D_k$, so $\mathbf{q}$ cannot lie in the interior of D_i : the pair (j, k) contributes no vertex. Each vertex on C_j then stems from the pair (i, j) and each on C_k from (i, k) , which gives $a_j = a_k = 1$ and $V = 4$. Otherwise $a_k \leq 1$ for all k and $V \leq 3$. For $V = 3$ every circle carries one arc and two vertices; two vertices from the same pair would leave the third circle without an arc, so each pair contributes one. For $V = 2$ the boundary consists of two arcs of the circles of one pair. A boundary without vertices is a full circle $C_i \subset D_j \cap D_k$, which the containment reduction has removed; therefore $V = 0$ implies $R = \emptyset$, and $V = 1$ does not occur. $\square$

The proposition replaces the 9, 27 and 16 cases of M. P. Fewell (2006), D. M. Kipping (2011) and T. A. Gordon & E. Agol (2023) by a count. Figure 1 shows the five topologies.

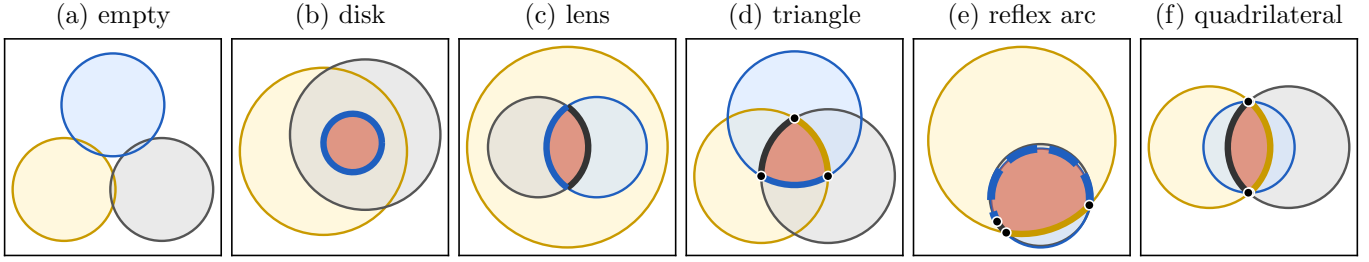


Figure 1. The topologies of the common overlap (red) of three disks after containment reduction. Vertices are marked by dots, boundary arcs are drawn in the colour of their circle, and the dashed arc in panel (e) exceeds π . Panel (a) shows three pairwise intersecting disks without a common point.

2.2. Boundary walk

Traversed counter-clockwise, ∂R runs counter-clockwise around the centre of each of its arcs, because R lies inside every disk. An arc of C_s between consecutive crossings on C_s has constant membership in the two other disks, and it belongs to ∂R exactly when its midpoint lies inside both. From every vertex $\mathbf{u}$ we therefore select the unique counter-clockwise arc to another vertex $\mathbf{v}$ on a shared circle C_s whose open span contains no other crossing of C_s and whose midpoint lies strictly inside the two other disks. With $\mathbf{u}' = \mathbf{u} - \mathbf{o}_s$ and $\mathbf{v}' = \mathbf{v} - \mathbf{o}_s$, the span test reduces to signs of cross products, and the midpoint is $\mathbf{o}_s \pm r_s(\mathbf{u}' + \mathbf{v}')/|\mathbf{u}' + \mathbf{v}'|$, with the positive sign when $\mathbf{u}' \times \mathbf{v}' > 0$. The successors form a single cycle through all vertices; any other outcome identifies a degenerate input. For $V = 3$ each circle carries exactly two vertices, the arcs follow the fixed adjacency of the three pairs, and the midpoint test alone fixes their orientation. For $V = 4$ the two arcs on the circle that carries all vertices join angularly adjacent vertices in counter-clockwise order. Both cases are evaluated without enumeration.

2.3. Area

Green's theorem gives $A = \frac{1}{2} \oint (x dy - y dx)$. For the arc of C_s from $\mathbf{u}$ to $\mathbf{v}$, parametrised as $\mathbf{o}_s + r_s(\cos t, \sin t)$, the identities $r_s(\sin \beta - \sin \alpha) = v_y - u_y$ and $r_s(\cos \beta - \cos \alpha) = v_x - u_x$ remove all trigonometric functions of the end angles:

$$A_{\text{arc}} = \frac{1}{2} [r_s^2 \Delta\theta + o_{s,x}(v_y - u_y) - o_{s,y}(v_x - u_x)], \quad (2)$$

$$\Delta\theta = \text{atan2}(\mathbf{u}' \times \mathbf{v}', \mathbf{u}' \cdot \mathbf{v}') \in (0, 2\pi]. \quad (3)$$

Equation (3) determines the arc from the geometry rather than from a chord length, so arcs longer than π require no separate treatment. Each term of Equation (2) equals the circular segment of the arc plus the signed triangle $(\mathbf{o}_s, \mathbf{u}, \mathbf{v})$, and the triangle terms combine around the cycle into the polygon of the vertices.

The area therefore is

$$A = \sum_{\text{arcs}} \frac{r_s^2}{2} (\Delta\theta - \sin \Delta\theta) + \frac{1}{2} \sum_{\text{edges}} (\mathbf{u} - \mathbf{p}) \times (\mathbf{v} - \mathbf{p}), \quad (4)$$

with $\mathbf{p}$ the first vertex. Equation (4) generalises the decomposition of [M. P. Fewell \(2006\)](#) to every topology and every arc length. It is evaluated in a form that contains no subtractive cancellation beyond that inherent in the data. For $\Delta\theta < 0.5$ the segment uses the Taylor series $\Delta\theta - \sin \Delta\theta = \sum_{n \geq 1} (-1)^{n+1} \Delta\theta^{2n+1} / (2n+1)!$, truncated after nine terms; for $\Delta\theta \geq 0.5$ it uses $r_s^2 \sin \Delta\theta = \mathbf{u}' \times \mathbf{v}'$, which requires no further transcendental function. The polygon is accumulated relative to $\mathbf{p}$, so that its rounding error scales with the size of R and not with the coordinates. The input is translated to the centre of the smallest disk. Figure 2 illustrates the walk and the decomposition.

2.4. Crossings and lenses

The crossings of C_1 and C_2 at distance d lie at distance $a_1 = [(d - r_2)(d + r_2) + r_1^2] / 2d$ from $\mathbf{o}_1$ along the line of centres and at distance h from it, where $4d^2 h^2$ is sixteen times the squared area of the triangle with sides (r_1, r_2, d) . We evaluate it with the ordering and bracketing of [W. Kahan \(2014\)](#): with $\alpha \geq \beta \geq \gamma$ the sorted sides,

$$4d^2 h^2 = [\alpha + (\beta + \gamma)][\gamma - (\alpha - \beta)][\gamma + (\alpha - \beta)][\alpha + (\beta - \gamma)], \quad (5)$$

which retains full relative accuracy for needle-like triangles, i.e., for nearly tangent circles. The lens follows from two segments with central angles $2 \text{atan2}(h, a_1)$ and $2 \text{atan2}(h, a_2)$, which replaces Equation (1).

2.5. Degenerate input

Tangencies and containments within 10^{-12} of the largest radius are resolved by the reduction of Section 2.1. A crossing within 10^{-9} of the third circle, a triple point, admits no reliable inside test; the solution is then evaluated with all radii shifted by $\pm 10^{-8}$ of the

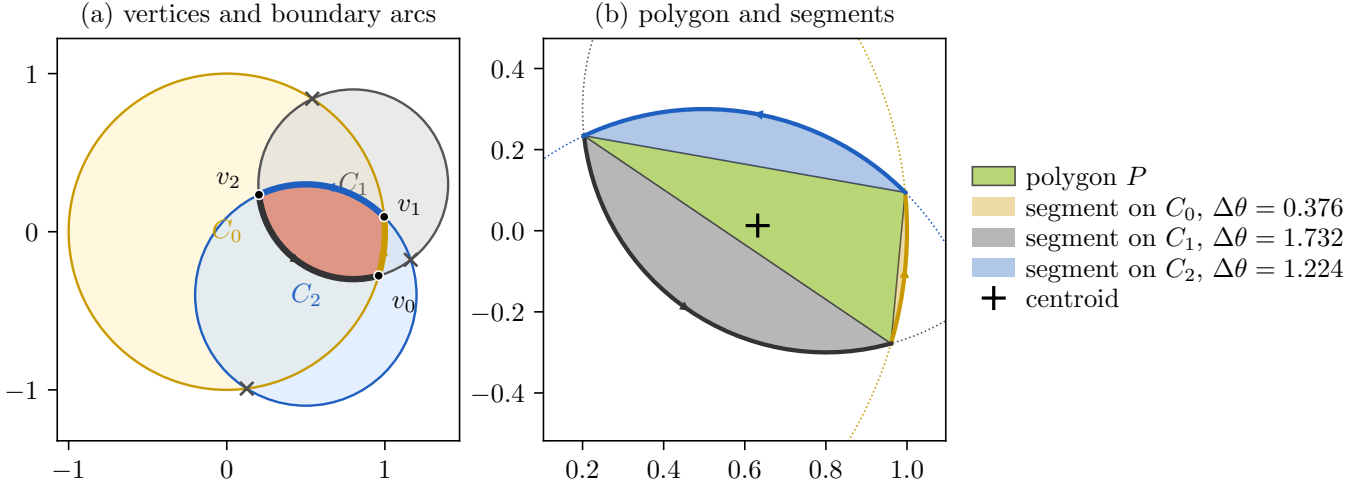


Figure 2. (a) Vertices of the overlap (dots), crossings outside the third disk (crosses) and the directed boundary arcs. (b) Decomposition of Equation (4): vertex polygon and one circular segment per arc, with the centroid of Equation (6).

largest radius and the two results are averaged. For a circle through both corners of a lens the averaged area agrees with the exact lens to 10^{-15} , and for three circles through one point it equals 7×10^{-18} .

2.6. Centroid

The decomposition of Equation (4) yields the first moments of R . The centroid of a segment with half-angle $\varphi = \Delta\theta/2$ lies on its axis, beyond the chord midpoint by

$$\begin{aligned} e &= r \left[\frac{4 \sin^3 \varphi}{3(2\varphi - \sin 2\varphi)} - \cos \varphi \right] \\ &= r \left(\frac{\varphi^2}{5} - \frac{13 \varphi^4}{1050} + \dots \right), \end{aligned} \quad (6)$$

and we evaluate the series, derived in exact rational arithmetic to order φ^{20} , for $\varphi < 0.5$. Together with the triangle fan of the polygon this gives the centroid of the overlap to within 1.7×10^{-12} of the smallest radius against an independent quadrature. The centroid provides the first-order correction for the variation of the stellar intensity across the overlap, which constant-intensity treatments of mutual events omit.

3. RESULTS

3.1. Accuracy

We measure accuracy against references computed with 50 significant digits (The [mpmath development team 2024](#)). The reference re-evaluates every vertex, angle, segment and polygon of Equation (4) in extended precision, with the topology taken from the double-precision classification; that classification agrees with an independent adaptive quadrature of the chord length

$\min_k(y_k + s_k) - \max_k(y_k - s_k)$, which contains no geometric case logic, to 3.2×10^{-14} of the smallest disk over 20,000 configurations. Five families sample the problem: general disks, disks of similar radius with strong overlap, a hierarchical star–planet–moon geometry with radii 1, 0.02–0.2 and 0.005–0.1, circular triangles produced by clipping a lens corner to a depth of 10^{-7} to 0.5 times the smallest radius, and thin lenses of a small disk at the rim of a large one.

The attainable accuracy is limited by the conditioning of the problem. We compute the componentwise relative condition number $\kappa = \sum_i |p_i| \partial A / \partial p_i / A$ over the nine input parameters by differentiation in extended precision. An algorithm whose error does not exceed a small multiple of κu , with $u = 2^{-53}$ the unit roundoff, returns the exact area of a configuration perturbed by a few units of roundoff in its data and is backward stable in the componentwise sense (N. J. Higham 2002). Table 1 and Figure 3 compare the present method with the direct evaluation of Equation (2) using the textbook crossings and Equation (1), with the corrected closed form of M. P. Fewell (2006) and T. A. Gordon & E. Agol (2023) for circular triangles, and with Equation (1) for lenses.

The error of the present method never exceeds $2.01 \kappa u$ in any of the 21,568 configurations; its median is $0.12 \kappa u$. The relative errors of 10^{-8} for the smallest triangles and 10^{-6} for the thinnest lenses are those of the data: a relative perturbation of 10^{-16} in the coordinates changes these areas by the same amounts. The direct evaluation of Equation (2) with the textbook crossings exceeds this limit by up to six orders of magnitude for small triangles, and Equation (1) by up to 16 orders of magnitude for thin lenses, where it returns areas wrong by factors up

Table 1. Maximum relative error and maximum error in units of κu

Family	N	This work	Eq. (2), textbook	Eq. (1)	Fewell + G&A
general	2155	$1.9 \times 10^{-13} / 1.1$	$2.0 \times 10^{-9} / 7.3 \times 10^2$	$2.0 \times 10^{-9} / 7.3 \times 10^2$	$1.5 \times 10^{-13} / 30$
similar	4995	$1.4 \times 10^{-14} / 2.0$	$2.2 \times 10^{-13} / 5.8$	$4.2 \times 10^{-13} / 9.0$	$4.0 \times 10^{-14} / 1.3 \times 10^2$
hierarchical	4418	$1.5 \times 10^{-12} / 1.3$	$2.1 \times 10^{-7} / 9.1 \times 10^4$	$2.1 \times 10^{-7} / 9.1 \times 10^4$	$3.4 \times 10^{-12} / 2.1 \times 10^2$
corner	5000	$9.9 \times 10^{-9} / 1.0$	$2.3 \times 10^{-2} / 1.1 \times 10^6$	...	$1.2 \times 10^{-8} / 1.4$
sliver lens	5000	$1.5 \times 10^{-6} / 0.9$	$3.6 \times 10^6 / 1.3 \times 10^{12}$	$1.4 \times 10^{11} / 7.8 \times 10^{16}$	...

NOTE— N is the number of configurations with non-zero area. Each entry gives the maximum relative error followed by the maximum of the error divided by κu . Equation (1) applies to lens results, the closed form of M. P. Fewell (2006) with the correction of T. A. Gordon & E. Agol (2023) to circular triangles.

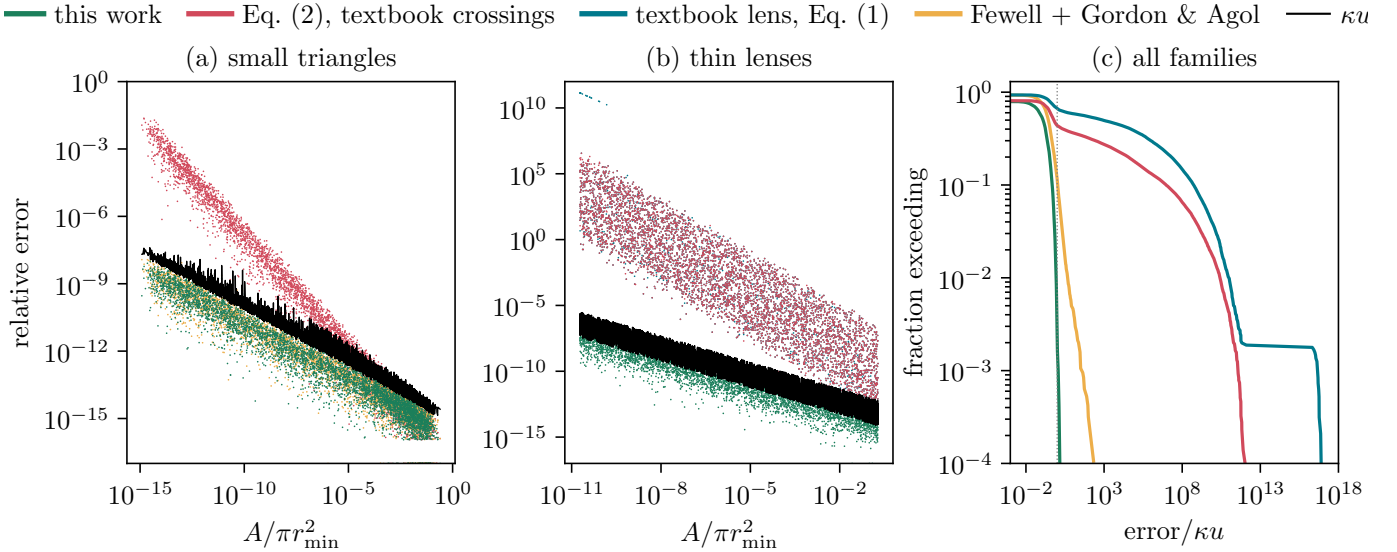


Figure 3. Accuracy against 50-digit references. (a) Relative error of small circular triangles versus their area in units of the smallest disk. (b) Relative error of thin lenses. The black curve marks κu . (c) Fraction of all 21,568 configurations whose error exceeds a given multiple of κu .

to 10^{11} . The corrected closed form of Fewell remains within $2 \times 10^2 \kappa u$ but covers circular triangles only.

3.2. Arcs longer than π

Among the circular triangles of Table 1, 101 contain an arc longer than π . The original Eq. 16 of M. P. Fewell (2006), adopted as Eq. 37 by D. M. Kipping (2011), errs on them by up to 66% of the area. Figure 4 shows one configuration: the original formula reduces the overlap from 0.859 to 0.654 of the smallest disk. A displacement of the smallest disk demonstrates that the original formula is continuous at the transition $\Delta\theta = \pi$ and diverges beyond it, which explains why the error escaped detection. The present method reproduces the corrected closed form to 2.4×10^{-12} of the smallest disk over all 4462 circular triangles without testing whether an arc is reflex.

3.3. Public software

We evaluated five public packages on the 40,000 configurations of Section 3.1, of which 21,568 have a non-zero overlap and 18,432 an empty one: **gefera** (commit e27e832, compiled from its Fortran source with its own flags and, with identical results, with `-O2`; T. A. Gordon & E. Agol 2023), **eulerr** 7.1.0 (J. Larsson 2026), **venn.js** (B. Frederickson 2018), **matplotlib-venn** 1.1.2 (K. Tretyakov 2025) and Shapely 2.1.2 with GEOS 3.14.1 (S. Gillies et al. 2025; GEOS contributors 2025) at 8, 64 and 512 polygon segments per quarter circle. **gefera** returns the flux of a star occulted by two bodies; for a uniform source this is the area of the union of the two shadows on the star, $|S \cap (P \cup M)|$, and we compare that quantity directly, which spares **gefera** the cancellation of an inclusion–exclusion. We count as a gross

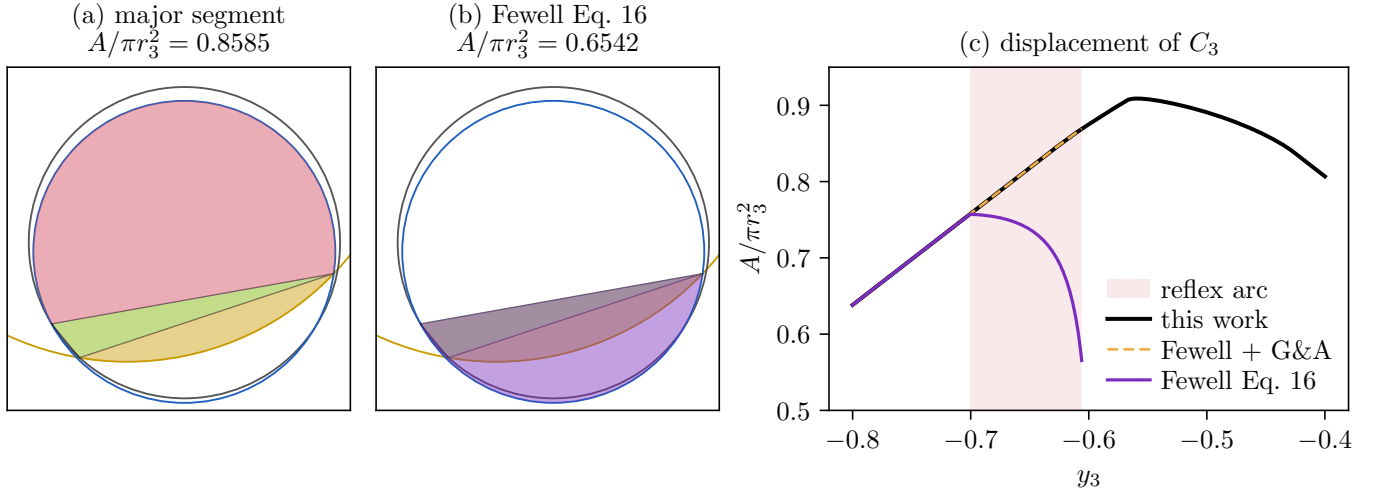


Figure 4. A circular triangle with an arc of 3.78 rad on the smallest circle C_3 . (a) Heron triangle and segments with the major segment on C_3 , which gives the exact area. (b) The segment of Fewell’s Eq. 16, which takes the minor arc. (c) Overlap area as C_3 moves along y : the shaded interval contains a reflex arc.

failure an undefined result, an exception, a relative error above 10^{-3} for a non-empty overlap, or a non-zero result exceeding 10^{-9} of the smallest disk for an empty one. Table 2 summarises the results.

gefera computes the union to a median relative error of 5×10^{-16} and to 2.3×10^{-10} in all hierarchical configurations, but it returns undefined values in all 31 configurations in which the disk of the planet lies inside that of the moon, and it assigns wrong areas, with errors up to 95%, to 90 circular quadrilaterals, 88 of which involve a body larger than the star. It also returns undefined values when a crossing of planet and moon lies exactly on the limb and when $b + r = 1$ for a single body. **eulerr** agrees with the reference in general configurations but fails on 310 small triangles and 448 thin lenses, and on 8 hierarchical configurations with bodies of radius 0.02 and 0.006, where it returns either zero or the full smallest disk in place of overlaps of 24 to 94% of that disk. **venn.js** fails on 1473 small triangles and 2216 thin lenses, with relative errors up to 1.6×10^6 : it evaluates circular segments from their width with an arccosine of $1 - w/r$ and tests containment with an absolute tolerance of 10^{-10} . **matplotlib-venn** raises an exception in 3746 configurations in which one disk lies inside another, because its region algebra excludes holes, returns zero in 886 configurations in which the smallest disk lies inside both others, and fails on 1017 overlaps smaller than 5×10^{-7} of the smallest disk. Shapely is limited by its polygon resolution: with 512 segments per quarter circle its median relative error is 1.9×10^{-5} , and small and thin overlaps are lost entirely.

3.4. Speed

We implemented the method in Python compiled with Numba (S. K. Lam et al. 2015) and timed batches of 2×10^4 configurations per topology on one core of an Intel Core Ultra 5 226V, taking the best of nine repetitions. The cost per configuration, including classification, translation and evaluation, depends on the topology. The representative mixture of all families costs 45 ns, dominated by the early exits of empty and contained configurations at 21–31 ns; circular triangles cost $0.17 \mu\text{s}$ and quadrilaterals $0.26 \mu\text{s}$. The specialised walks of Section 2.2 halve the cost relative to the general walk, which costs 117–427 ns. The stable evaluation of Equations (4)–(5) costs the same as the direct evaluation on the mixture and at most 41% more on lenses. The closed form of M. P. Fewell (2006) with the correction of T. A. Gordon & E. Agol (2023) costs $0.10 \mu\text{s}$ for a circular triangle when the topology is known in advance; it requires a case tree for all other configurations.

Table 3 and Figure 5 compare the public packages on the same core, each through its native interface, on the 40,000 test configurations and on two light-curve sequences of 2×10^5 positions of a planet and a moon on the stellar disk with fixed radii. The present method is 4–5 times faster than **gefera**, 6 times faster than **venn.js**, and 80 to 5700 times faster than **eulerr**, **matplotlib-venn** and Shapely.

4. DISCUSSION

The method separates topology from evaluation. The proposition reduces the topology to the number of vertices, and the walk determines the boundary without any formula that depends on the configuration. The evaluation then applies one expression to every topol-

Table 2. Public software on 40,000 configurations

Code	Undefined or exception	Wrong by $> 10^{-3}$	Max. error/ κu	Median error/ κu
This work	0	0	2.0	0.12
gefera , union area	31	90	...	...
eulerr 7.1.0	0	766	9.7×10^{13}	2.7
venn.js	0	3689	7.0×10^{13}	2.6
matplotlib-venn 1.1.2	3746	1903	4.5×10^{15}	10
Shapely, 8 segments	0	21568	7.0×10^{14}	1.1×10^{13}
Shapely, 64 segments	0	11022	4.6×10^{11}	2.1×10^{11}
Shapely, 512 segments	0	7294	7.1×10^9	3.5×10^9

NOTE—Errors relative to 50-digit references; κu as in Table 1. For **gefera** the union of the two shadows on the star is compared, whose condition number differs from that of the triple overlap; its maximum relative error is 0.95.

Table 3. Cost per configuration in ns

Code	Test set	$r_p, r_m = 0.1, 0.0092$	0.12, 0.05
This work (Numba)	82	47	99
gefera (Fortran)	...	202	514
venn.js (Node.js)	465	265	573
eulerr (R)	6825	5350	6550
matplotlib-venn	30074	45667	86749
Shapely, 8 segments	34761	35978	41859
Shapely, 64 segments	83172	92478	103402
Shapely, 512 segments	467650	907037	949028

NOTE—**gefera** accepts one pair of radii per call and is timed on the sequences only; it evaluates its limb-darkening integrals also for a uniform source. The **eulerr** times include $1.8 \mu\text{s}$ of R loop overhead per configuration; **venn.js** times include the construction of its circle objects. The test set contains more triangles than the mixture quoted in the text.

ogy. This structure removes the failure modes of earlier work at their origin: the reflex error of [M. P. Fewell \(2006\)](#) and [D. M. Kipping \(2011\)](#) cannot occur because no arc is inferred from a chord, and the precision loss of Equation (1) cannot occur because no arccosine of an argument near unity and no difference of nearly equal areas is formed. The resulting algorithm is backward stable in the componentwise sense over all tested geometries, including the near-tangent configurations that arise at every ingress and egress of a mutual event. The failures of the public packages in Section 3.3 fall into the same two classes: case trees that omit topologies, such as nested bodies and quadrilaterals in **gefera** or holes in **matplotlib-venn**, and a loss of the small and

thin overlaps of ingress and egress, which in **venn.js** stems from its segment formula and absolute tolerance and affects **eulerr** and **matplotlib-venn** alike.

The cost of the complete computation is that of a few square roots and one arctangent per boundary arc. In a photodynamical model the overlapping configurations form a small fraction of all samples, and most evaluations end in the early exits at 21–31 ns. The method therefore removes the three-disk geometry as a computational concern from light-curve models, and its error is bounded by the conditioning of the data.

The area and the centroid of the overlap determine the light blocked twice during a mutual event to first order in the gradient of the stellar intensity. The exact

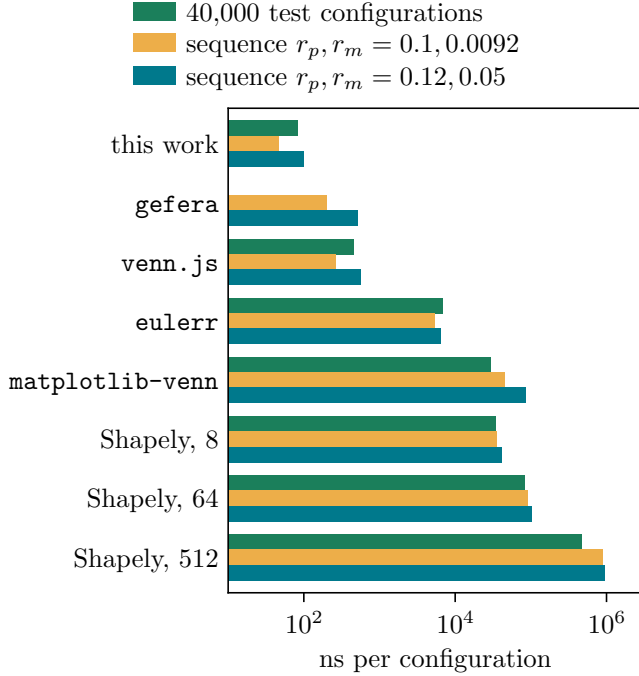


Figure 5. Cost per configuration on one CPU core (Table 3), logarithmic scale, for the 40,000 test configurations and two light-curve sequences of a planet and a moon with fixed radii. *gefera* accepts one pair of radii per call and is timed on the sequences only; Shapely is shown at 8, 64 and 512 segments per quarter circle.

limb-darkened flux requires the line integrals of A. Pál (2012) and T. A. Gordon & E. Agol (2023) along the same boundary arcs. The walk of Section 2.2 supplies these arcs, their end points and their orientation for every topology, and the stable crossings of Equation (5) supply accurate end points for the elliptic integrals; the boundary walk thus serves as a robust front end for exact mutual-transit photometry. The proposition and the

walk extend without change to disks of any relative size, including bodies larger than the star in stellar triples.

5. CONCLUSIONS

We have presented an exact solution for the area of the common overlap of three disks that is complete, backward stable and fast. A proof that the overlap has at most four corners reduces the geometry to five topologies resolved by one boundary walk. Green’s theorem, evaluated as circular segments with series for thin segments, a vertex polygon relative to its first vertex, and crossings from Kahan’s stable triangle area, yields errors below $2.01\kappa u$ for all 21,568 tested configurations, where the classical formulas err by factors up to 10^{16} of this bound or, for reflex arcs, by up to 66% of the area. Five public packages, including the mutual-transit code *gefera*, fail on between 121 and 21,568 of 40,000 configurations; the present method fails on none. The complete computation costs 45 ns per configuration on average and $0.26\mu s$ in the most complex topology, four to several thousand times less than the public packages. The same decomposition gives the centroid of the overlap in closed form. An open-source implementation is available at <https://github.com/user/repo>.

ACKNOWLEDGMENTS

AI (Anthropic Claude Opus 5.5) was used to derive the solution and assist the exposition of the text.

Software: NumPy (C. R. Harris et al. 2020), Numba (S. K. Lam et al. 2015), mpmath (The mpmath development team 2024), Matplotlib (J. D. Hunter 2007), *gefera* (T. A. Gordon & E. Agol 2023), *eulerr* (J. Larsson 2026), *venn.js* (B. Frederickson 2018), *matplotlib-venn* (K. Tretyakov 2025), Shapely (S. Gillies et al. 2025), GEOS (GEOS contributors 2025)

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